Branch Module

1. Objective

The Branch Module defines how an organization operates multiple independent branches within a single accounting system.
Each branch functions as a **self-contained operational unit** with its own transactions, accounts, balances, users, and reports, while still supporting organization-level consolidation.

2. Scope of the Branch Module

In-Scope

- Branch creation and lifecycle management
- Branch-level currency configuration
- Branch-wise association with transactions and accounts
- Branch-wise reporting (monthly, yearly, category-wise)
- User access control per branch (association only)

Out-of-Scope

- Organization-level master settings
- Transaction creation, approval, or posting
- Account balance calculation logic
- Global reporting rules
- Centralized audit policies (read-only consumption)

3. Branch Table

Column Name	Data Type	Nullable	Description
id	BIGINT UNSIGNED	No	Primary key
org_id	BIGINT UNSIGNED	No	Identifier to which branch belongs
name	VARCHAR(150)	No	Display name of branch
code	VARCHAR(150)	No	Org-scoped unique code
currency_code	CHAR(3)	No	Code associated with branch

country	VARCHAR(150)	Yes	Country where branch operates
status	ENUM('active','inactive')	No	Operational status
created_at	Timestamp	No	
updated_at	Timestamp	Yes	

- Primary key - id
- Foreign key - org_id
- Unique constraints - (org_id, code)
- Indexes - org_id

Dependent Modules

The following modules **depend on the Branch module** and reference `branches.id`:

Module	Dependency Type	Purpose
Accounts/Ledgers	FK reference	Branch-scoped accounts
Transactions	FK reference	Transaction ownership
User Branch Access	FK reference	Branch-level permission
Reports	Filter dimension	Branch-wise reporting
Audit Logs	Entity reference	Branch-related activity tracking

- Branch depends on Organization
- All financial and access data depend on Branch
- No child module may exist without a valid branch reference

4. Branch Entity – Functional Definition

4.1 Branch Creation

FR-BR-01

- The system shall allow authorized users to create a new branch.
- Mandatory fields:
  - Branch name
  - Branch code (unique per organization)

    - Country
    - Local currency
    - Timezone
- Default status shall be **Active**.

FR-BR-02

- Branch codes must be unique within the organization.
- Duplicate branch codes must be rejected.

4.2 Branch Update

FR-BR-03

- Authorized users shall be able to update:
    - Branch name
    - Status (Active / Inactive)
    - Address metadata (optional)
- Branch currency and country **must not be editable** after creation.

4.3 Branch Deactivation

FR-BR-04

- A branch can be marked **Inactive** only if:
    - No pending transaction approvals exist
    - No draft or submitted transactions exist
- Inactive branches:
    - Must remain available for historical reporting
    - Must not allow new transactions

4.4. Branch Configuration

Financial Configuration

FR-BR-05

- Each branch shall define:
    - Default local currency
    - Opening balance date
- All transactions created under the branch must inherit the branch currency.

4.5 Branch-Level User Access Control

1. Branch Access Mapping

FR-BR-08

- Users may be assigned access to one or more branches.
- Each assignment must include an access level:

Level	Description
View	Read-only-access
Editor	Allowed to create /edit data in the dependent module
Approver	Allowed to approve data in the dependent module
Branch Admin	Full control within branch scope

2. Access Enforcement

FR-BR-09

- The system must validate branch access before allowing:
    - Viewing transactions
    - Creating or editing records
    - Approving transactions
- Unauthorized access must be blocked at API and UI levels.

4.7. Branch Transaction Ownership

1. Transaction Association

FR-BR-10

- Every transaction must belong to exactly one branch.
- Branch ID must be mandatory for transaction creation.

2. Branch Immutability

FR-BR-11

- Branch ID cannot be modified once a transaction is created.
- Attempting to change branch ID must result in a validation error

4.8 Multi-Currency Handling (Branch Context)

1. Local Currency

FR-BR-12

- Transactions must be stored in the branch's local currency.
- Branch-level reports must default to local currency.

2. Base Currency (Optional)

FR-BR-13

- The system may store a converted base currency amount for consolidation.
- Base currency view must not affect branch-level calculations.

4.9. Audit and Logging

1. Activity Logging

FR-BR-14

- The system shall log all branch-related actions:
    - Creation
    - Updates
    - Transaction approvals
    - Status changes

2. Read-Only History

FR-BR-15

- Posted transactions must be read-only.
- Any correction must be done using reversal or adjustment entries.

4.10. Error Handling and Validation

1. Validation Rules

FR-BR-16

- The system must prevent:
    - Transactions without branch assignment

    - Use of inactive branches
    - Posting transactions outside an active financial year

4.11. Scalability and Performance

1. Scalability

FR-BR-17

- The system shall support:
    - Unlimited branches
    - Concurrent branch operations
    - Branch-level data isolation

5. Branch APIs

Below APIs manages operation-

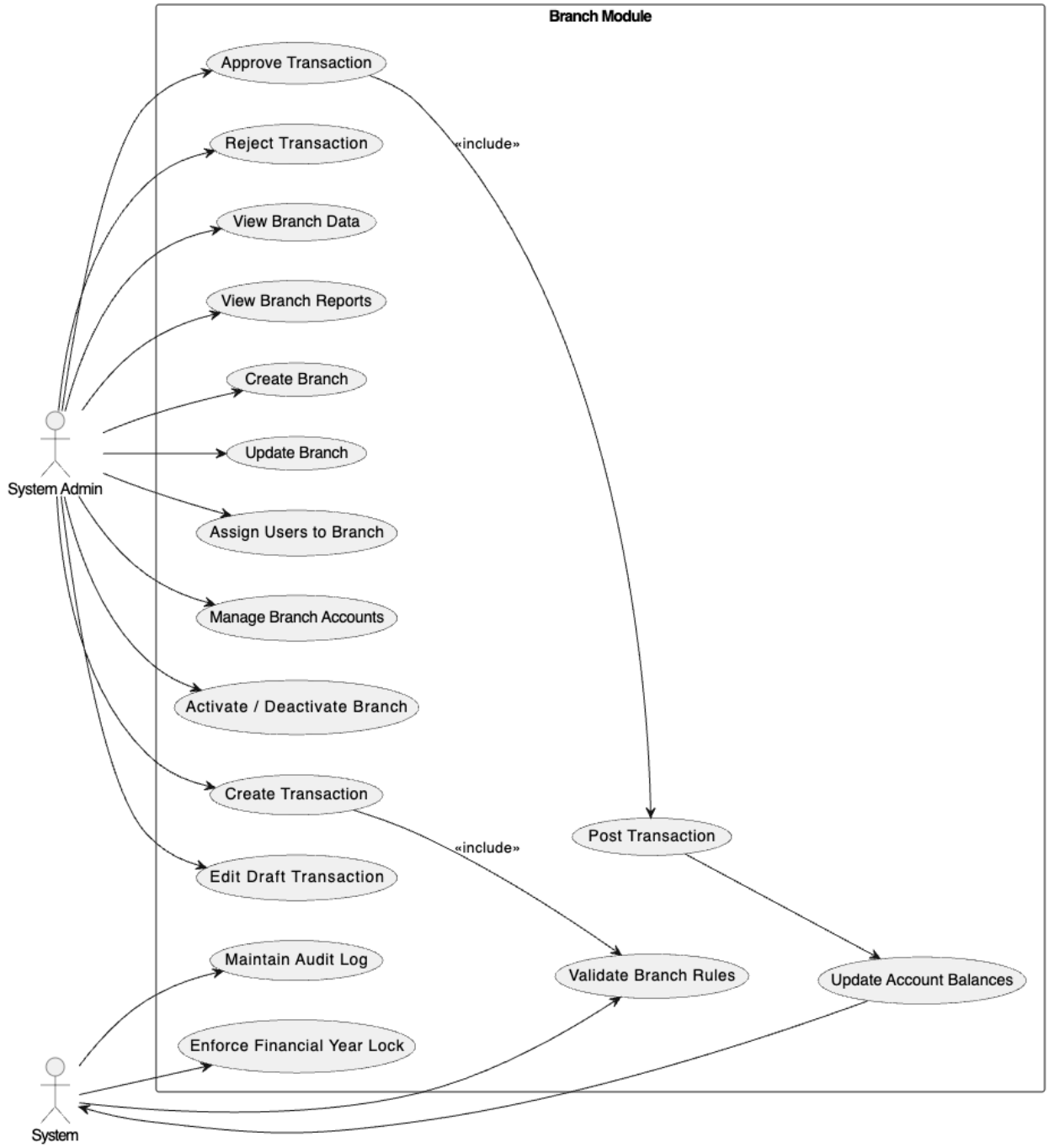
Item	Description
Method	Get
Endpoint	api/branches
Purpose	Retrieve all branches of the organization
Access level	Authenticated user
Input parameters	none
Output	List of branch records
Side Effect	None
Dependent Modules	Transactions, Accounts,RBAC, Reports
Notes	Includes active and non-active branches

Item	Description
Method	Post

Item	Description
Endpoint	api/branches
Purpose	Create a new branch under the organization
Access level	Admin
Input parameters	Branch name,Branch code, currency, country
Output	Newly created branch Id
Side Effect	Audit log entry
Dependent Modules	Transactions, Accounts
Notes	Branch code must be unique per organization

Item	Description
Method	Put
Endpoint	api/branches/:id
Purpose	Update branch master information
Access level	Admin
Input parameters	Branch Id
Output	Updated branch records
Side Effect	Audit log entry
Dependent Modules	Transactions, Reports
Notes	Branch Id and organization linkage are immutable

6. Use case


Branch Module
Approve Transaction
Reject Transaction
View Branch Data
View Branch Reports
Create Branch
Update Branch
Assign Users to Branch
Manage Branch Accounts
Activate / Deactivate Branch
Create Transaction
Edit Draft Transaction
Maintain Audit Log
Enforce Financial Year Lock
Post Transaction
Validate Branch Rules
Update Account Balances
«include»
«include»
System Admin
System